

Ideas Parked — Not Yet Merged (v14)

Staging document; each entry marks its eventual destination

Three rough sketches, offered together as a single pass at “what else might an atom’s own internal structure predict,” not as three independent proposals. None is formalized here. Each is checked against what the current paper already has — one turns out to restate a question already parked in v13; one turns out to be partly already true of an existing metric, in a way worth stating precisely rather than vaguely; one turns out to bear directly on an already-open gap, as a specific, structurally motivated special case of it. The sketches are kept in the user’s own words where precision would otherwise be lost, with the checking shown separately from the sketch itself.

1 Density, restated — not a new entry

“Density is number of total centers and info vectors (possibly weighted by sum vector magnitude).”

This is the same proposal as two conversations ago, in different words: the formula candidates are the same ones already weighed in ideas_v13 (N , D , S_{atom} , or S_{atom}/N), and “density” is the same word already retired once for γ_{atom} and recommended against for any of these. Nothing here changes that weighing. Recorded only so a future pass does not re-derive it: see ideas_v13, Sections 3–5, for the full accounting and the standing recommendation (keep N/D as mass-number/atomic-number; leave “density” retired; park S_{atom}/N rather than name it).

2 Energy as shell distribution — partly already true, worth stating precisely

“Energy is the distribution of strongest vectors... in upper shells vs lower shells. If an information atom has more vectors in lower shells and less in higher shells and another has more vectors in higher shells and less in lower shells then ones with higher shells carry more energy.”

Checked: this is already true of $\mathcal{E}_{\text{atom}}$, for a reason worth stating rather than leaving implicit. $\mathcal{E}_{\text{atom}} = \sum r_i^2$ already rewards concentration toward high r over an even spread, for the same total strength — confirmed directly: four vectors summing to $S_{\text{atom}} = 2.0$ give $\mathcal{E}_{\text{atom}} = 1.0$ spread evenly (0.5, 0.5, 0.5, 0.5), and $\mathcal{E}_{\text{atom}} = 1.81$ concentrated toward the extremes (0.05, 0.05, 0.95, 0.95) — an 81% increase for the identical total strength, driven entirely by where that strength sits. This is not a coincidence of the example: squaring is convex, so for any fixed sum, spreading values apart (in particular, pushing some toward the high end) can only raise the sum of squares, never lower it. σ_{atom} , checked on the same pair, tracks the same shift (0 spread evenly, 0.333 concentrated) — confirming $\mathcal{E}_{\text{atom}}$ and σ_{atom} are already telling a connected story, not two unrelated ones.

What is not yet true, and would need its own definition. σ_{atom} is explicitly stated elsewhere in the paper to be *blind to which side* concentration sits on — it measures evenness versus unevenness, not high-shell versus low-shell. Checked directly: nothing currently distinguishes “three vectors low, one high” from “three vectors high, one low” when their evenness happens to match, the way the sketch’s own two-atom comparison needs. $\mathcal{E}_{\text{atom}}$ partly closes this on its own, by weight of the convexity argument above, but it is a single summed number, not an explicit shell-indexed quantity; whether that is enough, or whether a genuinely new, shell-position-weighted quantity (something like a sum of r_i^2 each multiplied by its own shell index) earns its own name under the ideas_v13 rule, is not decided here. A promising direction, not a finished one.

3 Matter and antimatter — a specific, structurally motivated case of an already-open gap

“If info atoms have all the vectors in right vertical sphere then it is matter and another in left half then it is antimatter particle with potential to cancel out all the information.”

Checked: this bears directly on Gap 1, not on new ground. The paper already has one proven cancellation: a mark and an equal-strength anti-phase mark, same direction $\hat{n}$, Bundle as two separate seats ($\oplus$ does not cancel them) and are only cancelled on a later reading, by Mix, into black (“Pathogen cancellation,” a costume of Axioms 4 and 5). That is opposition in χ at one shared direction — a narrow, well-defined special case, not a general rule for differing directions.

The sketch proposes a second, geometrically distinct kind of opposition: not same- $\hat{n}$, opposite- χ , but opposite- $\hat{n}$ itself — antipodal points on the sphere ($\hat{n}$ against $-\hat{n}$). This is exactly the question “Open: Mix across differing directions” (Gap 1) leaves unsettled, and the leading candidate on file for that gap, Route 1, argues for refusing combination across differing directions *in general*, on evidence from the **chladni** inclusion. The sketch is not in conflict with Route 1 so much as it is the same kind of move the existing cancellation rule already made: Route 1 is about the generic case (some unprivileged angle apart), while both the existing same-direction cancellation and this sketch’s antipodal cancellation are specific, maximally symmetric cases — 0° apart in χ reversed, and 180° apart in $\hat{n}$ exactly — each singular enough on the sphere to plausibly warrant its own rule rather than inheriting the generic refusal.

What would be needed before this is more than a sketch. A stated rule for what Mix (or some other named reading) does at exactly $\hat{n}_B = -\hat{n}_A$, checked the way the existing cancellation was checked — does it reduce to black the way same-direction opposite-phase does, or to something else; does it require equal strength, the way the existing case implicitly does (an equal-strength anti-phase mark); and whether “cancels out all the information” means the pair reduces to the Null State of Axiom 2 specifically, or some other outcome not yet named. None of this is decided here — the sketch is recorded as a specific, well-motivated candidate for Gap 1’s hardest symmetric case, not as a solved instance of it.

4 Status

All three sketches are the user’s own framing, offered as a rough pass with the explicit caveat that it is not yet known which will hold up. Section 1 is closed by reference to prior work, not by new reasoning. Sections 2 and 3 are each partly verified (the numbers and the citations above

are checked against the current paper) and partly open (the shell-weighted formula in Section 2; the exact cancellation rule in Section 3), and both are left open rather than pushed toward a formalization the sketches themselves do not yet claim to have earned.